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Comparison of Single-Arm and Dual-Arm Collaborative Robots in Precision Assembly

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Abstract: The aim of the study is a multi-criteria comparative evaluation of robots co-operating with humans in single- and dual-arm variants used for the process of precise assembly of complex parts. RobotStudio simulation software with the Signal Analyzer add-on was used for comparative analyses. These studies were conducted as case studies. A robotic station was designed for the assembly of a computer motherboard and two robot variants were programmed to perform the assembly task while maintaining the same motion parameters and functions for both. Then, the TCP motion trajectories associated with the robot were analyzed, as well as monitoring signals from the robot controller during simulation, such as time, speed, acceleration and energy consumption. The costs and profitability of the robot variants were also calculated. The percentage share of tasks performed in the process was also analyzed, divided into assembly tasks and free movements. The differences between the robots in this process include time, 21 s single-arm versus 14 s dual-arm robots. The main influence on achieving the programmed speed was the length of the robot's TCP motion path. In most cases, the maximum programmed speed of 200 mm/s was achieved. The single-arm robot proved to be more energy-efficient, but the dual-arm robot proved to be 20% faster, which in the long run proved to be a more profitable robot. The profitability of the dual-arm robot paid off after eight months of operation. The case study presented in this paper, assembling a computer motherboard using single- and dual-arm collaborative robots, provides a guide for conducting similar comparative analyses of different robotic stations. Simulations enabled reliable verification of collaborative robots in technological processes, supporting the design of production processes and the analysis of several variants of robotic solutions.

Keywords: industrial robots; collaborative robots; assembly; precision engineering

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1. Introduction

Robotization of technological processes is one of the aspects of Industry 4.0. Progressive competition and a dynamically changing market result in the need for robotization of production processes, which translates into increased efficiency, precision and quality of manufactured products [1]. Robotization of processes is of particular importance in many industries: automotive, furniture, metal processing, electronics and medicine [2]. Most applications of industrial robots concern tasks related to object manipulation (pick and place), as well as welding, gluing, palletizing [3,4], painting and quality assurance [5,6]. All of these operations can be part of the assembly process, which can be robotized.

Statistical data from robot suppliers and robotics associations consolidated by the IFR Statistical Department indicate that the electronics industry has been the main recipient of

industrial robots since 2020. In 2021, the number of installed robots increased by 24%, and in 2022 it increased by 10%. A growing trend is the use of collaborative robots (cobots) in electronics assembly processes. The market share of collaborative robots, which include the ABB YuMi robots, increased by 31%. Since 2017, the demand for robots in the electronics industry has increased by an average of 5% per year. The overall long-term growth trend is expected to continue [7]. Statistics provide grounds for research on the theoretical and practical use of collaborative robots, especially those used in electronic parts assembly processes.

The implementation of robotization and the selection of its most advantageous solution are based on selection criteria such as costs and time of the manufacturing process. To assess which variant is the most advantageous (for the selected evaluation criterion), the station or robotic line is modeled and simulated in a virtual environment. In such an environment, manipulation [8–10], grinding [11], palletizing [12], painting, milling, welding [13], spraying [14], sealing [15] and 3D printing [16] operations can be simulated. The authors of these publications indicate the benefits of offline robot programming, mainly checking the program code without any interruptions in the ongoing process, assessing the path and movement parameters of the tool or gripper and, above all, the ability to analyze the time and costs of the programmed solution. Borboni et al. discussed the advantages of simulating the work of collaborative robots in precise and expensive manufacturing processes to best reflect the real process [17].

One of the popular collaborative robots is ABB's YuMi, with an ergonomic shape modeled on human arms. The robot has 14 axes, 7 on each arm. This robot is equipped with many safety elements, which makes it possible to eliminate expensive security measures such as barriers, curtains and fences and allows work in the immediate vicinity of people. Chemweno et al. developed dynamic safety zones to activate the robot to stop at a user-defined distance from an object [18]. The ABB YuMi collaborative robot allows for the effective implementation of assembly processes where other industrial robots require special equipment and advanced grippers. The accuracy and repeatability of positioning at the level of 0.02 mm [19] allows the robot to be used in precise and repeatable assembly processes [20]. Using the YuMi robot, it is possible to simulate the technological process using one or two arms, which is impossible in the case of standard industrial robots of one kinematic structure. Precision engineering is an important element of modern manufacturing processes affecting quality, efficiency and innovation [21].

The interest in collaborative robots in terms of identifying research and development opportunities, as well as manufacturing applicability is one of the most important topics of publications in recent years. Due to the relatively short period of application of collaborative robots (since 2008), many research possibilities are recognized as not previously described in the literature [22]. Collaborative robot research focuses mainly on increasing the humanization of the robot and more effective cooperation with humans, the development of Industry 5.0, and above all, case studies of their applications in specific manufacturing processes [23–25]. The studies about cooperation between humans and robots include technological safety [26], artificial intelligence methods for optimizing tool paths [27], the structure and mechanism of operation of collaborative robots [28] and cost-optimal approaches [29]. Michalik et al. proposed a vision system for recognizing human hand gestures, based on artificial intelligence methods, for more efficient cooperation with collaborative robots. The proposed approach turned out to be 90% effective, i.e., the robot recognized 9 out of 10 gestures presented by the human. The research results are promising, but to improve the system's performance, it would be necessary to work on the stability of lighting conditions and the distance of the hand from the camera [30]. Eyam et al. developed a method for recognizing a person's emotional state using electroencephalography

(EEG) to adjust the parameters of a collaborative robot's work, mainly the speed of the robot's movements, to ultimately improve work comfort. Despite interesting research results, the application of this method in practice can be cumbersome due to the nature of EEG, which is a non-invasive but tactile method [31]. Pang et al. developed a flexible robot skin made of piezoresistive nanocomposite with similar tactile perception to human skin to enhance the naturalness and safety of human–robot interaction [32]. Cohen et al. developed vocal communication between a robot and human [33]. Human-centered studies and personalization of production are the main assumptions of Industry 5.0. Therefore, scientific research in the area of collaborative robots and human–robot interaction is consistent with current trends in industrial development. Zafar et al. concluded that collaborative robots are not only designed to perform tasks, but according to the idea of Industry 5.0, they are also to enhance human potential. Therefore, these robots do not only serve physical assistance but also go beyond this framework. Cobots equipped with systems based on artificial intelligence are supposed to increase human creativity and decision-making to improve production processes [34]. Dhanda et al. described that the use of contemporary scientific achievements in manufacturing processes entails interdisciplinary challenges from the fields of robotics, manufacturing, psychology and ethics [35]. Despite the many challenges associated with human–robot interaction, case studies of their use in specific applications provide practical knowledge. Giberti et al. described the general application of collaborative robots in industrial processes [24]. Lunt et al. designed a modular station of three collaborative robots performing multi-task laboratory work, a fully robotic process of powder X-ray diffraction [36]. Mozafari et al. described a station using a two-arm ABB YuMi robot coupled with a vision system to control the quality of the manufactured food product. The robot's arm movements were selected to imitate those of a human [37]. Li et al. investigated the possibilities of increasing the accuracy of these robots to increase the precision of work [38]. Polverini et al. appreciated the kinematic redundancy and lightweight design of collaborative robots for small parts assembly processes. The authors described an example of assembly of a plastic pipette using the ABB YuMi dual-arm robot [39]. Bejarano et al. designed and experimentally validated the YuMi robot for part assembly. These studies confirm the assumption that the design results are consistent with the experimental ones [40]. Pantanetti et al. presented a case study using YuMi robots in the assembly of mechanical components. The authors assume that these robots maintain standards of precision, safety and energy efficiency. However, they do not provide detailed calculations in their publication [41]. Polonara et al. presented a case study of manufacturing and assembling automotive parts using collaborative robots. The authors encourage further research on the implementation of collaborative robots in assembly processes, especially more complex and precise tasks [42].

These studies meet the current trends described in publications from recent years. The case studies of collaborative robots have a practical contribution to the design of manufacturing processes with the inclusion of human–robot interactions. This work compares single-arm and dual-arm robots for the purpose of multi-criteria evaluation of both solutions. This research presents the results of simulations of the operation of two variants of robots collaborating in the assembly process along with the multi-criteria analysis of time, acceleration, speed, energy consumption and costs. The key here is to analyze the precision assembly processes of complex parts using collaborative robots. In the literature, a multi-criteria comparison of single- and dual-arm robots in a precision assembly process has not yet been presented as a case study. Therefore, this study may constitute a guide for conducting similar comparative analyses of various robotic stations.

2. Materials and Methods

2.1. Characteristics of the Assembled Parts

The computer motherboard is one of the most important parts of the computer, on which other components are mounted. Its task is to enable communication between individual components, control the flow of energy and data and ensure the stability and reliability of the entire system. An example computer motherboard is presented in Figure 1. Assembling the motherboard is a key step in building a computer. The essence of this process is the proper connection and protection of the processor and RAM chips on the motherboard to ensure the proper functioning of the computer [43]. Computer motherboards have different types of connectors to connect many components. One of the most important connectors is the processor slot to which the processor is mounted (Figure 1). Another important connector is the RAM memory slot, where RAM modules are installed, enabling fast data processing (Figure 1). Precise assembly of the computer motherboard is possible due to the high accuracy and repeatability of modern industrial robots. The vision system associated with the ABB YuMi collaborative robot (ABB Ltd., Zurich, Switzerland) is used to locate components and perform precise and correct assembly. The maximum jaw spacing is larger than the overall dimensions of the motherboard components that are mounted. Due to the fact that this robot is equipped with force sensors in the grippers, the risk of crushing computer motherboard components is eliminated. The arrangement of elements for assembly is shown in Figure 1.

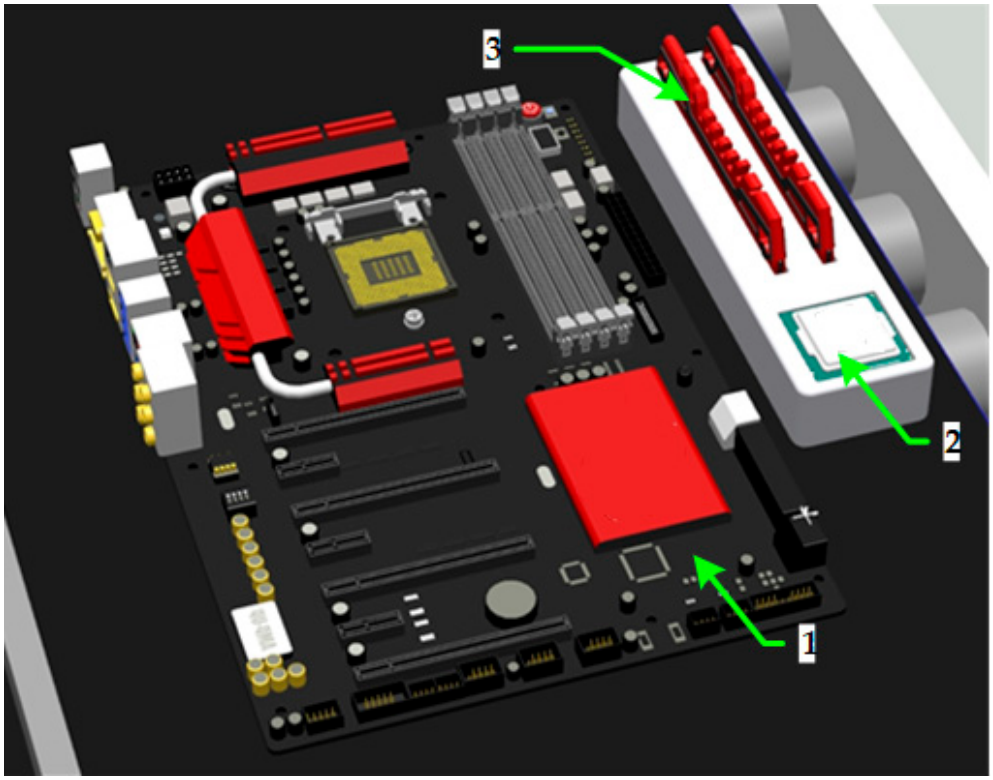


Figure 1. Assembled parts. 1—Computer motherboard, 2—Processor, 3—RAM module.

2.2. Robotic Cell

A comparative evaluation of two variants (Figure 2) of collaborative robots for the assembly task was carried out in ABB's RobotStudio. The most important functions and capabilities of the RobotStudio environment used for the analysis are:

- construction and 3D visualization of a robotic station allowing one to observe how the station works and how the robot moves during the assembly process before implementation in the real environment,
- offline programming of a collaborative robot means creating a program, as well as its editing, testing and optimization without the requirement of physically working with the robot,
- communication between the robot, external and peripheral devices and sensors makes it possible to conduct a full simulation of the entire production station and also enables subsequent implementation of ready-made programs to all devices, not only the robot,
- a library of functions in which many ready-made solutions can be used to quickly create simulations, including smart components for dynamic conversion of station inputs and outputs,
- simulation of the robot and objects of the entire station in order to test and optimize the robot's movement trajectory, but also additional results of the robot's work such as collisions, speeds, accelerations, energy consumption and many others,
- a realistic user interface for offline programming that significantly mirrors online programming, for example a virtual operator panel (FlexPendant).



(**a**) (**b**)

Figure 2. Collaborative robots: single-arm ABB YuMi IRB 14050 (**a**) and dual-arm ABB YuMi IRB 14000 (**b**) [44].

In these studies, two collaborative robots are compared in single- and dual-arm versions. Figure 2 shows the compared ABB YuMi IRB 14000 and ABB YuMi IRB 14050 collaborative robots with a lifting capacity of 0.5 kg per arm with a dedicated 0.2 kg gripper attached.

Both robots are equipped with grippers with one servo module (Figure 3). The grippers are integrated with vision and vacuum to increase the precision of the tasks performed.



Figure 3. The gripper of the YuMi ABB collaborative robot [19].

The station was equipped with a conveyor belt and a plastic container designed specifically for the robotic production line used to assemble parts on the computer motherboard using the ABB YuMi collaborative robot. The view of the station with one and two-arm robot is shown in Figures 4 and 5.

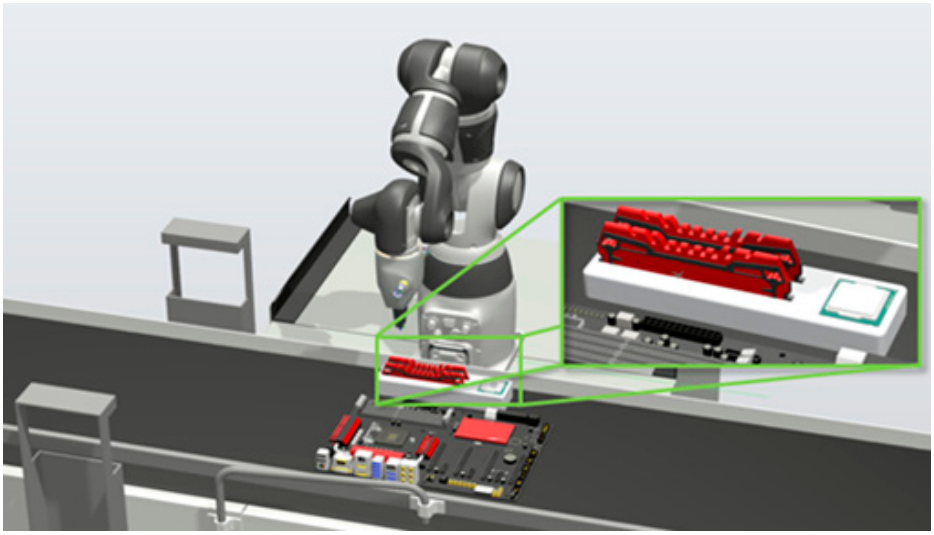


Figure 4. View of the station with a single-arm robot.

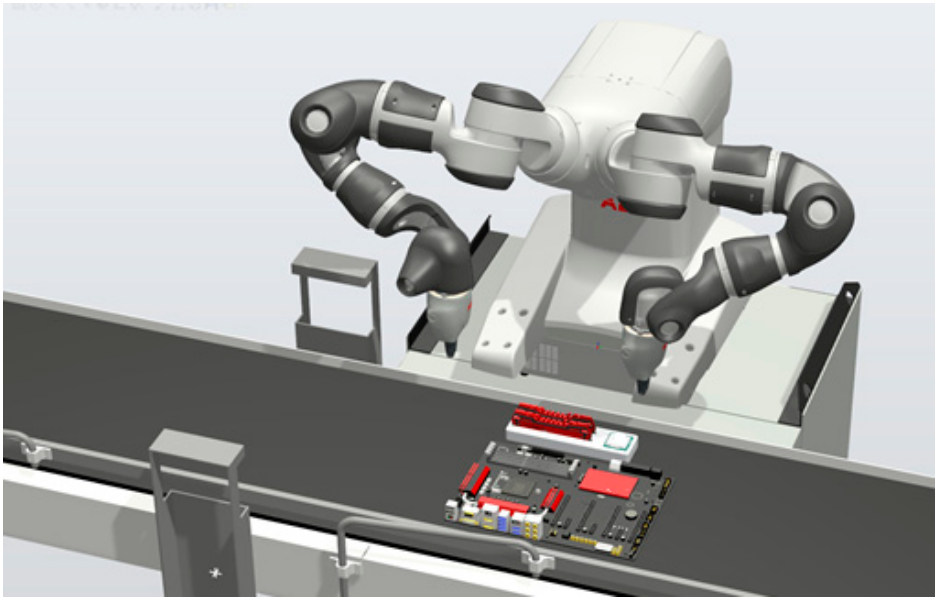


Figure 5. View of the station with a single-arm robot.

The single-arm robot takes and presses RAM number 1 first, then RAM number 2 and finally the processor. The dual-arm robot arms perform RAM assembly at the same time. The left arm takes RAM No. 1 and the right arm No. 2. Then, the left arm mounts the processor and the right one at the same time presses the memory in the sockets. Synchronization of the movements of the left and right arms can be achieved by programming each arm separately, taking into account the collision detection function, so that any user errors do not cause a system failure. In this approach, each of the robot arms is treated as a separate robot but controlled by the same controller. The recommended solution is the MultiMove function and the RAPID SyncMoveOn instruction, which allows for coordinated movement of both robot arms. The flowchart of tasks performed by robots is shown in Figure 6. Figure 7 (single-arm robot) and Figure 8 (dual-arm robot) show the

TCP trajectory of the robot for the assembly task using the built-in TCP Trace function, which is a module designed to visualize the tool center point (TCP) motion trajectory [8].

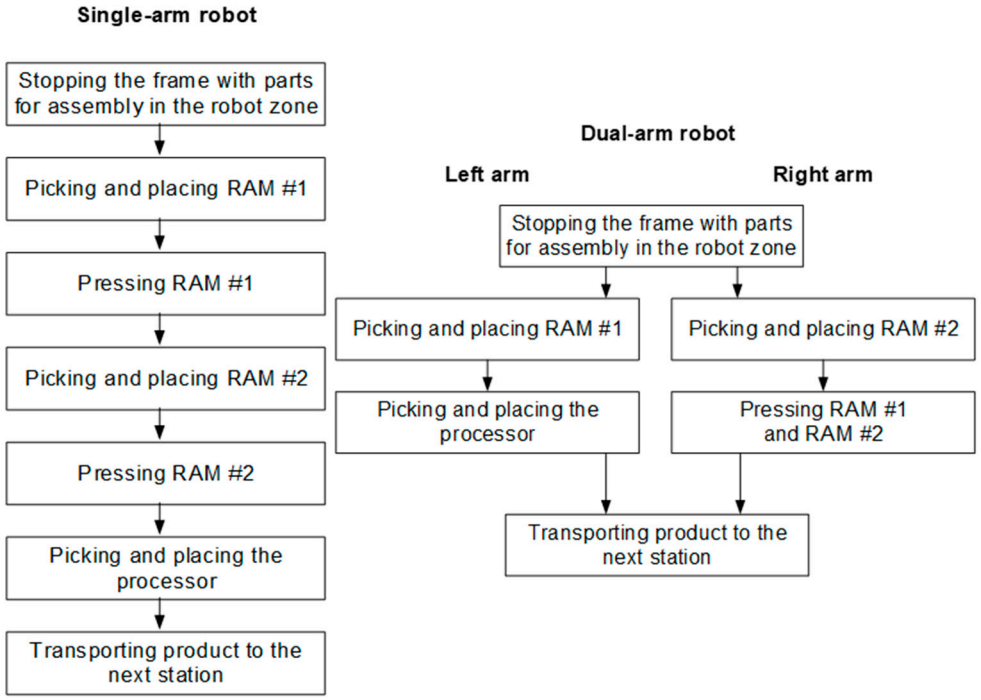


Figure 6. Flowchart of tasks performed by robots.

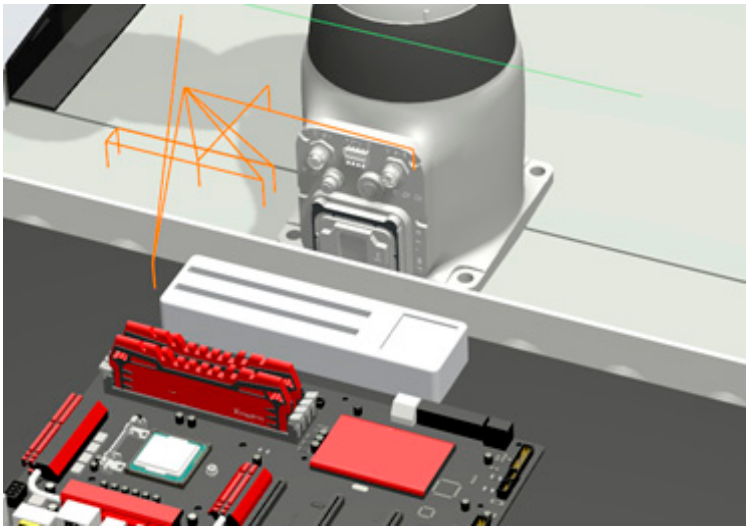


Figure 7. Motion trajectory for a single-arm robot.

Figure 9 shows the outline of the motion trajectory for a dual-arm robot, and Figure 10 shows the trajectory for a single-arm robot. The following designations are adopted: 1—starting position of the arm, 2—RAM memory acquisition position, 3—RAM memory installation position, 4—RAM memory clamp position, 5—processor acquisition position, 6—processor installation position.

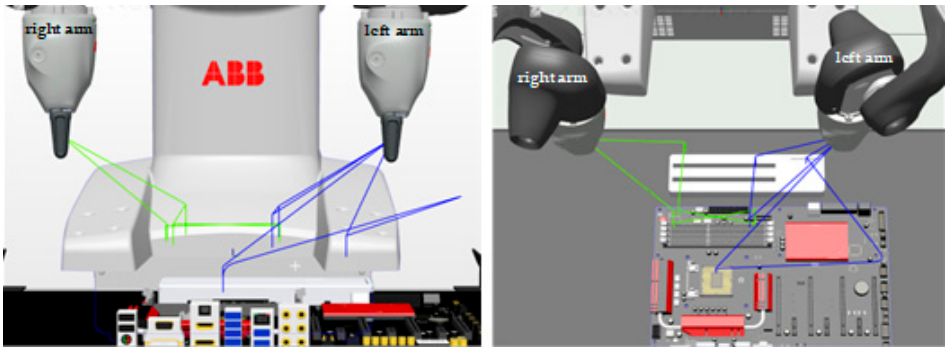


Figure 8. Motion trajectory for a dual-arm robot.

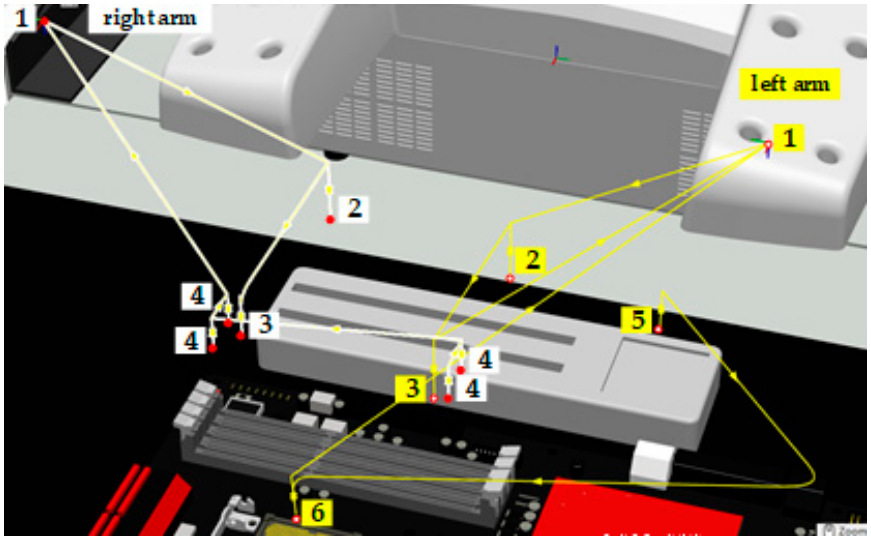


Figure 9. Outline of the trajectory of a dual-arm robot.

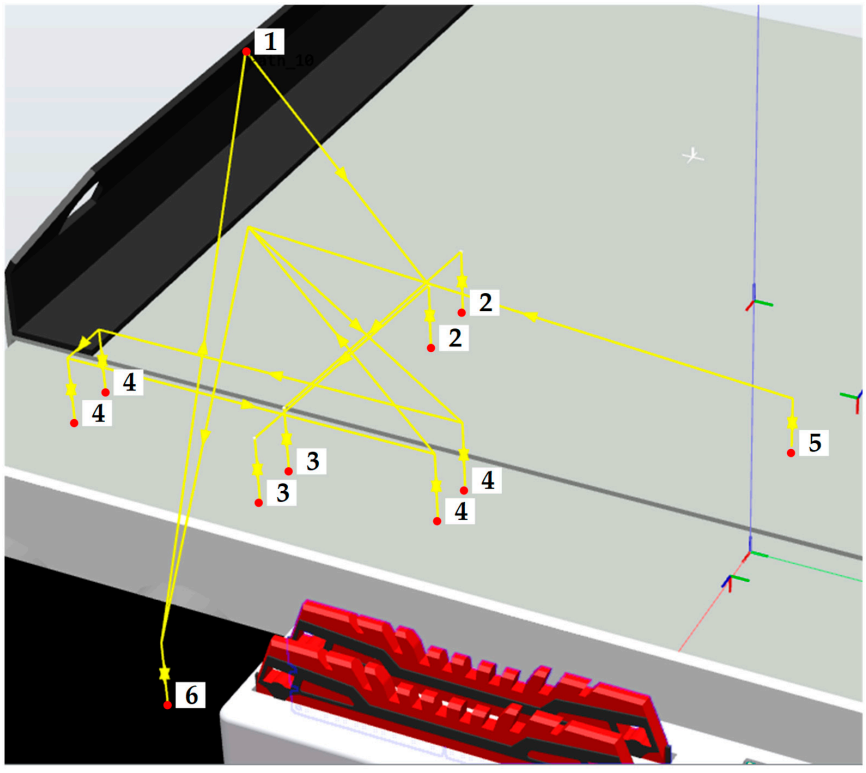


Figure 10. Outline of the trajectory of a single-arm robot.

In these studies, avoiding collisions between the robot arms and station objects was achieved at the stage of planning/programming the robot arms' movement path. The positions were corrected during program testing so that collisions did not occur. The RobotStudio program has the Create Collision Set function, which indicates with which objects a collision should be detected by changing color, making it easier to correct the program when simulating work. The second option to avoid collisions is the Collision Free Path function, which is used to design a path avoiding declared objects. The appearance of the robot arms near the boundaries of a given zone causes it to stop. It is also possible to declare appropriate signals depending on the robot's position. The program is compiled so that the robot arm operates in a given zone when the signal changes its value (work permission).

2.3. Rapid Program Code

The robot program was written taking into account the elimination of collisions between the arms in the dual-arm robot variant. For obvious reasons, this problem does not occur in the case of a single-arm robot. The robot motion parameters and functions used during testing were the same for both the single- and dual-arm robots. Accurate movement was used in places where parts were collected and placed and with a speed of 50 mm/s. The maximum movement speed for each path was 200 mm/s. The movement between the programmed points of the tool path was in straight lines. The torque and acceleration values were not modified. The default values for both robots were used.

The RAPID (Robot Application Programming Interface Description) code used in these studies is a programming language related to the RobotStudio software, which is a programming environment for ABB industrial robots. RAPID code written in RobotStudio can be compiled and sent to a real robot to control its operation. Table 1 shows sample instructions used in the program representing previously declared motion parameters for the planned motion trajectory of single- and dual-arm robots. Robot trajectory simulations are presented in the Supplementary Materials.

Table 1. Explanation of functions in the RAPID code.

Function	Explanation
CONST robtarget Home	Assigning coordinates and names for the point
MoveL Home, v200, z0	Linear movement command to a point called "Home" with a speed of 200 mm/s and zone z0
WaitTime 1;	Command to wait the robot for a specified time
Reset/Set PickRam	Setting the signal that controls the movement of the mounting frame in position 0/1

2.4. Methods of Comparative Evaluation of Collaborative Robots

The evaluation criteria are the process time, the speed and acceleration of the robot's TCP movement and energy consumption. Additionally, assuming, among others, the value of profit, man-hour, purchase or energy cost, and by calculating efficiency, it is possible to analyze the costs of implementing single- and dual-arm robots in the production process of computer motherboards. However, the results are universal and can also constitute suggestions and a basis for choosing a robot variant for other assembly processes.

Time, speed, acceleration and energy consumption analyses were performed based on simulation results in RobotStudio. RobotStudio Signal Analyzer was used for this purpose, which is an application for monitoring signals from the robot controller during simulation. This is particularly useful for multi-criteria analysis of designed robotic processes. The same motion parameters and functions were maintained for the single- and dual-arm robots. The

description of the functions used in the form of pseudocode is given in Table 1. The signal analyzer calculates not only energy consumption but also time, speed and acceleration based on the created pseudocode.

3. Results and Discussion

3.1. Time of the Assembly Process

In variant I, a single-arm YuMi robot was used to assemble the computer motherboard. The total simulation time is 20.6 s. During the simulation, the robot performs all necessary tasks related to motherboard assembly, such as picking up components and placing them on the board. The times of individual tasks performed by the robot are presented in Table 2 and in the chart in Figure 11.

Table 2. The time needed to complete individual assembly tasks by a single-arm robot.

Task	RAM 1	RAM 2	Pressing the RAMs	Processor	Free Movements	Total
Time [s]	2.1	2	3.6	4.2	8.7	20.6

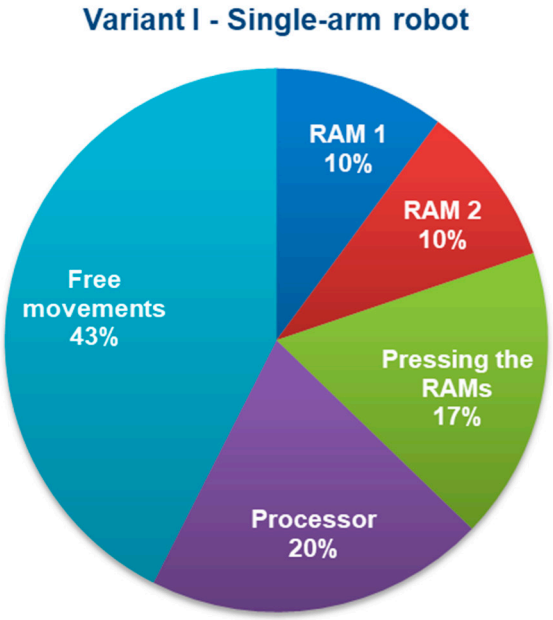


Figure 11. Percentage of individual robot assembly tasks for variant I.

The largest part of the time involved free movements within the workstation, mainly extending the gripper to the place where the part is picked up. These movements accounted for 43% of the time of the entire assembly process. This indicates that the robot performed activities related to the manipulation of elements for 8.7 s. The next longest task was mounting the processor due to the longest distance the robot arm had to move. The operation of pressing the RAMs took 3.6 s, or about 17% of the time of the entire operation, and mounting the RAMs was 10% for each of them.

In variant II, a dual-arm YuMi robot was used. The total simulation time is 13.6 s. Both robot arms are able to perform tasks simultaneously, which speeds up the motherboard assembly process. During the simulation, one robotic arm can pick up parts while the other places them on the board. All values related to time analysis are presented in Table 3 and charts in Figures 12 and 13. In the time analysis in variant II, the percentage of time of each task on both arms was considered and, additionally, as in variant I, the ratio of the robot's work to free movements.

Table 3. The time needed to perform individual assembly tasks by each arm of a dual-arm robot.

Task	RAM 1	RAM 2	Pressing the RAMs	Processor	Free Movement	Total
Left arm Time [s]	2.6	-	-	3.9	7.1	13.6
Right arm Time [s]	-	2.4	2.1	-	9.1	13.6

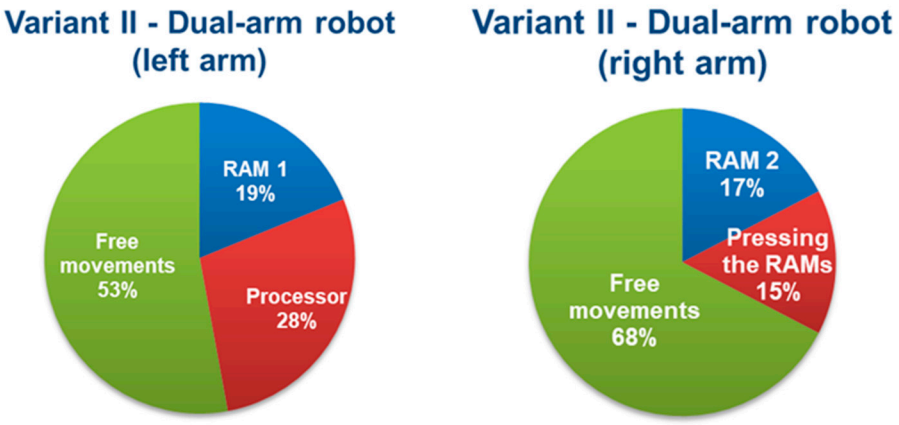


Figure 12. Percentage of individual assembly tasks for each robot arm for variant II.

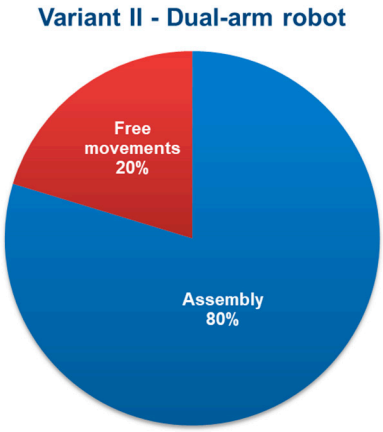


Figure 13. Percentage of tasks for variant II—dual-arm robot.

The operating time of the left robot arm in variant II includes a large amount of free movements of over 50%. This arm places the first RAM in its appropriate slot and then reaches for the processor and mounts it in the designated place. These activities take 2.6 s and 3.9 s respectively. For the right arm, the free movements are much higher and amounts to 68%. This arm is responsible for mounting the second RAM and for pressing RAM No. 1 and 2. Both of these tasks took a total of 4.5 s. Taking into account the entire operation performed by both robot arms in variant II, summing up the assembly times of each part, free movements took only 20% of the time of the entire process.

3.2. Speed and Acceleration

The sequence of movements of both compared robots (single-arm and dual-arm) in relation to time is presented in Table 4.

Table 4. Robot movement sequences.

Single-Arm Robot		Dual-Arm Robot			
		Left Arm		Right Arm	
Time [s]	Sequence	Time [s]	Sequence	Time [s]	Sequence
0	start	0	start	0	start
0.72	free movement	0.768	free movement	2.784	free movement
1.08	working movement	1.344	working movement	3.36	working movement
2.112	picking RAM1				
2.76	free movement	2.4	picking RAM2	4.416	picking RAM1
3.144	working movement	3.48	free movement	5.544	free movement
4.176	inserting RAM1	4.056	working movement	5.832	working movement
4.92	free movement				
5.328	working movement	5.112	inserting RAM2	6.864	inserting RAM1
6.36	picking RAM2	7.584	free movement	8.352	free movement
7.032	free movement	7.8	working movement	8.64	working
7.416	working movement	8.832	picking processor		movement
8.448	inserting RAM2	11.04	free movement	8.664	pressing RAM 1 (corner 1)
9.912	free movement	11.208	working movement	8.904	free movement
10.272	working movement			9.168	working movement
10.32	pressing RAM 1 (corner 1)	12.24	inserting processor		
				9.192	pressing RAM 1 (corner 2)
11.064	free movement	13.632	free movement	9.84	free movement
11.448	working movement			10.128	working movement
11.496	pressing RAM 1 (corner 2)				
				10.152	pressing RAM 2 (corner 1)
11.784	free movement			10.392	free movement
12.168	working movement			10.656	working movement
12.216	pressing RAM 2 (corner 1)				
				10.68	pressing RAM 2 (corner 2)
12.936	free movement			13.632	free movement
13.344	working movement				
13.368	pressing RAM 2 (corner 2)				
15.24	free movement				
15.552	working movement				
16.608	picking processor				
18.192	free movement				
18.552	working movement				
19.584	inserting processor				
20.568	free movement				

The course of the actual TCP speeds of the gripper of a single-arm robot and two arms of a dual-arm robot is shown in Figure 14. The programmed speeds may differ from the real ones due to the kinematics of the robot or the length of the movement path and the inability to achieve the programmed speed on a given section of the path.

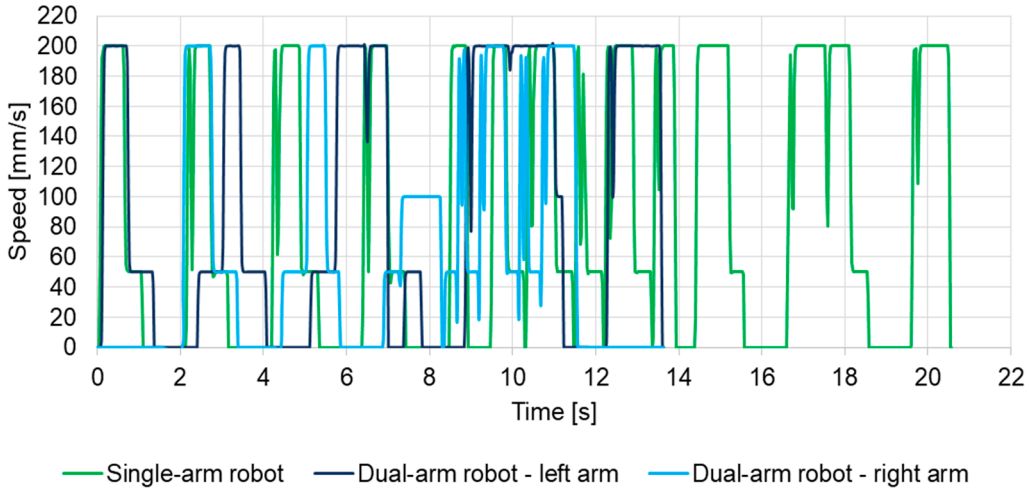


Figure 14. TCP speeds versus time.

ABB YuMi robots have no difficulty in achieving a maximum speed of 200 mm/s on programmed paths. It is not achieved only when pressing the RAM bone with the right arm of the robot because the section of the path between individual points is too short and it is not possible to develop the programmed speed. The speed of the TCP of the gripper to the position of picking parts from the container and placing them in the computer motherboard is reduced to 50 mm/s to minimize the risk of collisions and possible displacements of moved assembly parts. Figure 15 shows the accelerations achieved by the gripper's TCP during the process.

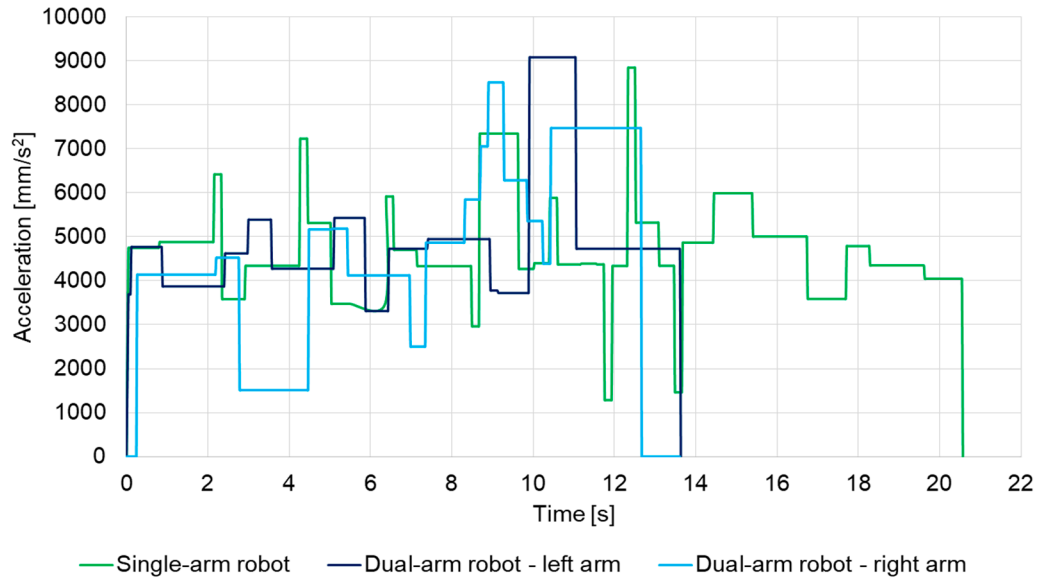


Figure 15. TCP accelerations versus time.

The graph of the acceleration of the robot's arms at the TCP point versus time suggests that the largest acceleration peaks for all arms occur between 8 and 12 s. For a single-arm robot, it is the movement it performs over the processor in preparation for lifting it, and for a dual-arm robot, it is the movement associated with moving the processor to the appropriate place. The upward trends in most cases are related to starts after stops in the parts pick-up and put-down zones. Downward trends appear as the gripper approaches the target points where direct assembly is performed.

3.3. Energy Consumption

Industrial robots are an integral part of modern production lines, contributing to increased efficiency and precision of processes. However, their growing importance requires taking into account energy consumptions, because energy is one of the key factors influencing the costs and sustainable development of the industry. In this case, two stations were analyzed—with a single- and dual-arm YuMi robot. These results are presented in Table 5 and in the graph in Figure 16.

Table 5. Energy consumption.

	Time [s]	Energy Consumption [J]
Variant I—single-arm robot	20.6	205.8
Variant II—dual-arm robot	13.6	260

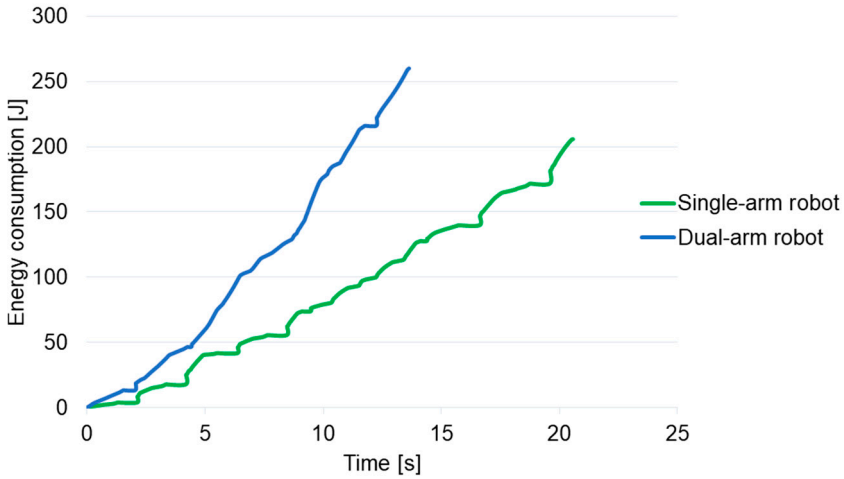


Figure 16. Energy consumption.

During one assembly cycle, a single-arm robot consumes 205.8 J of energy to power the robot in 20.6 s. Whereas a dual-arm robot consumes 260 J in 13.6 s. Based on these results, it can be concluded that the variant I single-arm robot is more energy-efficient. However, a two-arm robot performs the assembly operation faster and using 20% more energy, which could turn out to be a much more financially profitable option in the case of large-scale production. When comparing collaborative robots, it should be noted that operating time is not proportional to energy consumption.

3.4. Costs

In the comparative analysis of the costs of two variants of collaborative robots, two points of view are taken into account: the profit and loss account from the application of robots in assembly, as well as the unit cost of robotic assembly.

In order to assess which robot would be more profitable during production, a cost estimate was prepared. The first expense that must be incurred when implementing a robotic station is the cost of purchasing the robot. This cost estimate does not include the purchase prices for the stand's equipment, as it would be the same in both variants. The second important aspect is the efficiency of robots, i.e., the number of products they can assemble per month. The robots are located next to the assembly line and it was necessary to take into account the time during which the assembled frame with parts moves to the place where the process will start. The value of this allowance was set at 5 s and cycle times of 25.6 s for a single-arm robot and 18.6 s for a dual-arm robot. This cost estimate assumes that the robot will work in three shifts for 22 days a month (average value of working

days). Based on these data, the results are presented in Table 6. The man-hour values were adopted on the basis of the possibility of producing a given number of products per hour, cost of purchasing the robot, system maintenance and costs of an employee monitoring work. The monthly profit is calculated based on the Equation (1):

$$\begin{gathered}\text{Monthly profit} = \\ (\text{No. of cycles per month} * \text{Cost of producing 1 product}) - (\text{No. of shifts} * \\ \text{No. of hours per shift} * \text{No. of working days per month} * \text{cost per man} \\ \text{hour}) - (\text{Energy consumed in a month} * \text{cost per kWh})\end{gathered} \quad (1)$$

Table 6. Cost and profit estimation of the robotic assembly (estimated data for 2024) [45,46].

	Single-Arm Robot	Dual-Arm Robot	Unit
Purchase cost	37,369 [47]	61,968 [48]	€
Cycle time	25.60	18.60	s
Energy consumed per cycle	206	260	J
Number of shifts	3	3	
Number of hours per shift	8	8	
Number of cycles per hour	141	194	pcs
Energy consumed per hour	28,941	50,323	J
Energy consumed per number of shifts	694,575	1,207,742	J
Number of working days/month	22	22	
Energy consumed in a month	15,280,650	26,570,323	J
Energy consumed in a month	4.24	7.38	kWh
Number of cycles per month	74,250	10,2194	pcs
Energy consumption in the year	51	89	kWh
Number of cycles per year	891,000	1,226,323	pcs
Cost per man hour	10.78	21.55	€
Cost of production pcs.	0.32	0.32	€
Cost per kWh	0.27	0.27	€
Monthly profit	18,312	21,655	€
Unit cost	0.047	0.058	€/pcs.

It is important in cost analysis to determine the unit cost. Using division calculation, which is one of the basic ways of determining unit costs and calculated as the quotient of the sum of production costs and the production volume—Equation (2):

$$K_j = \frac{K}{P} \quad (2)$$

where:

K_j—unit cost,
K—total costs (purchase cost, energy, labor costs, etc.),
P—production volume.

Taking into account the data in Table 6, the unit cost for a single-arm robot is 0.047 €/pcs., and for a dual-arm robot it is 0.058 €/pcs.

For the analyzed assembly process, robots bring profit, and their implementation would pay off after about 3 months. A two-arm robot is a more advantageous economic solution due to higher profits per month. Cumulative profit is the difference between the financial results (profit/loss) of the dual-arm robot and the financial results (profit/loss) of the single-arm robot, taking into account subsequent months. In this way, it can be seen that from the perspective of time, the dual-arm robot brings more benefits. Comparing the profits for both solutions and assuming that we allocate all the profit to cover the purchase costs of the robots, it can be seen that the solution with the dual-arm robot will yield more profit after eight months than the solution with the cheaper single-arm robot. Therefore, in the long run, it is beneficial to use a dual-arm robot. Figure 17 shows a graph of the cumulative profit of a dual-arm robot versus a single-arm robot.

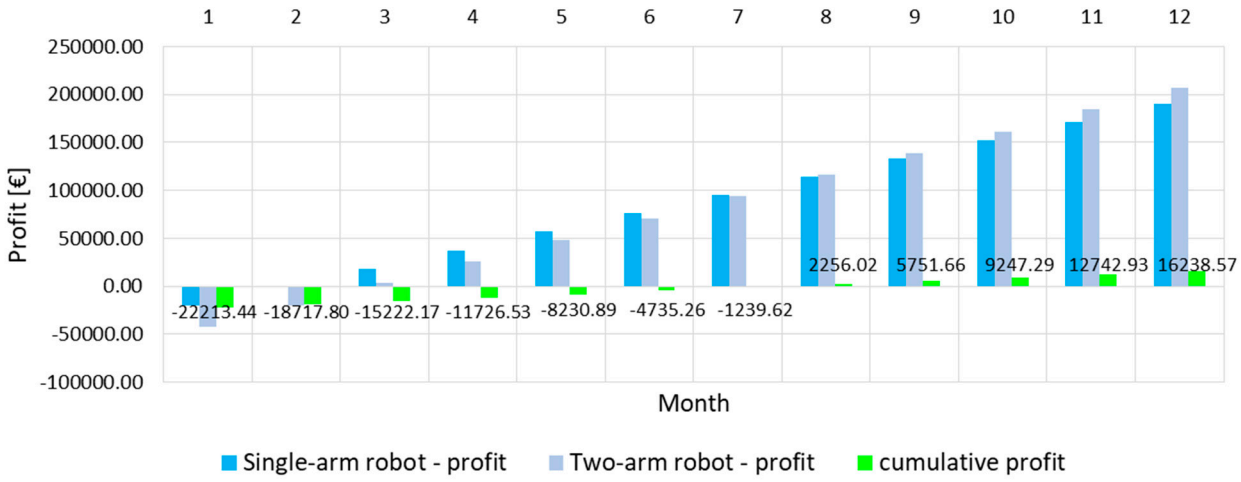


Figure 17. The cumulative profit of a dual-arm robot versus a single-arm robot.

4. Conclusions

Collaborative robots are becoming more and more widely used in assembly processes due to their precision of movements and the possibility of installation in the direct work zone of humans. Comparative research on collaborative robots is part of the current trend in the industry of using precision robots that can cooperate with humans and are additionally equipped with vision.

Dual-arm robots can simultaneously perform various activities assigned to each of the arms separately. It is important to eliminate potential collisions that can occur not only with station equipment but also between arms.

Analyzing the work of collaborative robots in a virtual simulation environment has many benefits, including, most importantly, monitoring information about the movement trajectory, collisions, process time, energy consumption and movement speed. Offline programming capabilities facilitate the procedure of building robotic stations, as well as selecting robots and robot movement parameters without interruptions in the ongoing production process.

The differences between the two robot variants include time: 14 s in favor of the dual-arm robot and 21 s as a disadvantage of the single-arm robot. The time is influenced by the speed of TCP movement, with the achievement of the programmed speed depending on the length of the robot's TCP path. In most cases, the maximum programmed speed of 200 mm/s was achieved. The choice of speed also depends on the precision of the task being performed, and it was reduced to 50 mm/s during the strict assembly task. The single-arm robot proved to be more energy-efficient, but due to the more favorable process execution time of the dual-arm robot, it turned out to be a more cost-effective robot in the long run.

During one assembly cycle of the analyzed process, a single-arm robot consumes 205.8 J of energy in 20.6 s. whereas a dual-arm robot consumes 260 J in 13.6 s. Based on these results, it can be concluded that a single-arm robot is more energy efficient. However, the two-arm robot performs the assembly operation faster and using 20% more energy, which could prove more profitable in the case of large-scale production.

Analyzing the cost values, it can be concluded that both robots are profitable, and their implementation would pay off after about three months. The cost-effectiveness of the dual-arm robot paid off after eight months of operation.

The obtained results are universal, and the simulation performed in a similar way allows for a comparison of collaborative robots in other technological processes. These studies can demonstrate a procedure for evaluating industrial robots in an offline pro-

gramming mode, suggesting a way in which other robotic manufacturing processes can be verified. In addition, these studies present an interesting case study of computer motherboard assembly, from the design stage of the robotic station, through robot programming and comparative analysis of both solutions, time, speed, acceleration, energy consumption and costs.

The simulation results were obtained from three tests. The virtual environment is distinguished by ideal conditions without disruptions and thus the repeatability of results. Therefore, the compared criteria of both robot variants, such as time, speed, acceleration and energy consumption, were characterized by stability. The manufacturer states that the simulated work in the virtual environment corresponds to that in reality, especially for new and regularly inspected robots. Implementation of the pseudocode developed in the virtual environment in the real robot station should therefore lead to obtaining the same results. The limitations of virtual environments are the lack of consideration of random disruptions or variable conditions, including robot adaptability, maintenance requirements or effective human–robot interaction. Simulations have the advantage of enabling a reliable way to verify collaborative robots in technological processes to support the design of production processes and analysis of several variants of robotic solutions.

Supplementary Materials: The following supporting information can be downloaded at: https://www.mdpi.com/article/10.3390/app15062976/s1, Video S1: RobotStudio dual arm. Video S2: RobotStudio single arm.

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